Week 4 — Intelligent Systems Assignment Summary

Task B: Cleaning Performance

The cleaning pipeline was evaluated using Mean Absolute Error (MAE). A lower MAE indicates that the estimated temperature is closer to the reference temperature.

Dataset	Readings	MAE (°C)
Raw sensor	254	0.582
Edge-cleaned	242	0.001
Cloud-cleaned	288	0.014

The raw sensor data contains missing readings and faulty values. Edge cleaning removes suspicious readings using lightweight checks, while cloud cleaning can use a longer time history to reconstruct missing values. The results show the trade-off between removing unreliable readings and retaining a more complete dataset.

Task C: Edge vs Cloud Cleaning

Cleaning step	Location	Reason
Temperature range check	Edge	Simple and fast; avoids transmitting impossible readings.
Stuck-sensor detection	Edge	Uses a small recent buffer and reduces unnecessary network traffic.
Sudden-change detection	Edge	Lightweight checks can flag suspicious changes before transmission.
Gap filling	Cloud	Can use readings before and after a gap and has more memory for historical data.

Task E: Privacy

Temperature patterns may indirectly reveal when people enter or leave a room. I would aggregate readings into 15-minute averages at the edge device and avoid sending unnecessary exact timestamps or occupancy-related information. This reduces detailed data transmission and may improve privacy while also reducing network

traffic. However, aggregation can hide short-term changes and reduce prediction accuracy. Therefore, the system should retain only the time and temperature detail necessary for reliable temperature control.